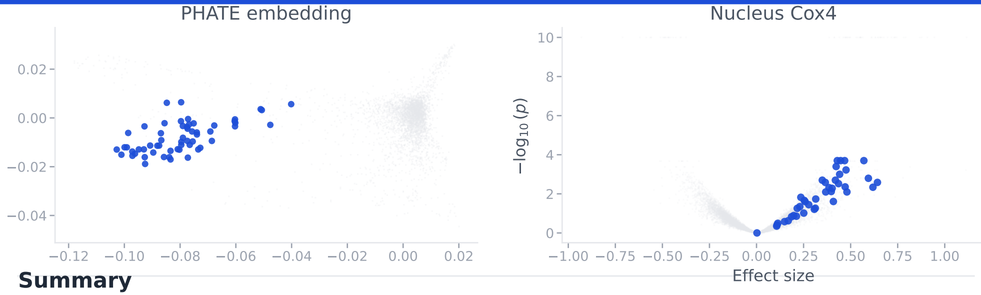
Ribosome biogenesis and small ribosomal subunit (40S) assembly

Cluster 2



Summary

This cluster shows HIGH confidence for ribosome biogenesis and 40S small ribosomal subunit assembly. Over 75% of genes are established components: 22 ribosomal proteins (RPS family), 13 SSU processome/pre-rRNA processing factors (UTP, BMS1, PNO1, etc.), 5 translation initiation factors (EIF3 complex), and supporting factors (POLR1B for rRNA transcription, XPO1 for nuclear export, RIOK2 for late ma...

Established genes

RPS20, RPS8, RPS7, RPS12, RPS13, RPS23, RPS19, RPS10, RPS24, RPS9, RPS3A, RPS14, RPS26, RPS2, RPS15, RPS15A, RPS17, RPS5, RPS21, RPS11, RPSA, FAU, PNO1, UTP11, BMS1, PDCD11, BYSL, RCL1, HEATR1, LTV1, UTP15, UTP20, DHX37, DDX21, UTP14A, BUD23, NOP14, WDR43, EIF3B, EIF3A, EIF3CL, EIF3G, EIF3C, POLR1B, RIOK2, AATF, XPO1

Novel & uncharacterized genes

BBIP1 *novel*

BBSome component required for primary cilia assembly and microtubule stability. While unexpected in this cluster, some ribosomal proteins have ciliary functions. Connection may be indirect through

DDX18 *novel*

RNA helicase with established roles in R-loop resolution and pluripotency maintenance. While it has some connection to nucleolar function and rDNA loci, its role in ribosome biogenesis is partially

DDX10 *novel*

RNA helicase primarily characterized in innate immunity and inflammasome activation. No established role in ribosome biogenesis, but DDX family helicases are known to function in RNA processing.

SRFBP1 *novel*

Involved in transcriptional regulation and SLC2A4 processing. Limited characterization overall. Potential role in regulating ribosomal gene transcription, but function remains unclear.

PPRC1 *novel*

Coactivator for mitochondrial biogenesis and CREB/NRF1 targets. While it regulates transcription related to cell growth, no direct link to ribosome biogenesis is established. May coordinate

ANAPC2 *novel*

Core component of APC/C ubiquitin ligase controlling cell cycle progression. No established role in ribosome biogenesis. May regulate stability of ribosome biogenesis factors or coordinate ribosome

ZCCHC9 *novel*

Known as transcriptional regulator (NF-kappa-B, serum response element), but limited functional characterization overall. No established role in ribosome biogenesis. May regulate transcription of

RPS10-NUDT3 *unc.*

Fusion gene or readthrough transcript with minimal characterization. RPS10 component is established in ribosome function, but the fusion product's role is unclear.

+ 3 more

47 established · 8 novel · 3 uncharacterized